

By the definition of $p_i^{(m)}$, for $m \geq 3$

$$\begin{aligned}\mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-1)} \right] &= \frac{1}{N'} \mathbb{E} \left[y_{i,j}^{(0)} \sum_{k=1}^N y_{i,k}^{(0)} \right] + \frac{1}{N'} \mathbb{E} \left[y_{i,j}^{(0)} \sum_{k=1}^n y_{i,k}^{(m-2)} \right] \\ &= \frac{p_i + (N-1)p_i^2}{N'} + \frac{n}{N'} \mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-2)} \right]\end{aligned}$$

and

$$\mathbb{E} \left[y_{i,j}^{(0)} p_i^{(1)} \right] = \frac{1}{N} p_i + \left(1 - \frac{1}{N} \right) p_i^2 = \frac{p_i + (N-1)p_i^2}{N}$$

which gives

$$\mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-1)} \right] = \frac{1 - \left(\frac{n}{N'} \right)^{m-2}}{1 - \frac{n}{N'}} \frac{p_i + (N-1)p_i^2}{N'} + \left(\frac{n}{N'} \right)^{m-2} \mathbb{E} \left[y_{i,j}^{(0)} p_i^{(1)} \right].$$

Setting $\alpha := n/N'$, we have $N = (1 - \alpha)N'$ and hence

$$\mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-1)} \right] = \frac{p_i + (N-1)p_i^2}{N}$$

for all $m \geq 2$. Plugging this back to the expression (15) for $v_i^{(m)}$ gives

$$v_i^{(m)} = \frac{(1 - \alpha)(1 + 2\alpha)}{N} p_i + \left(1 - \frac{1}{N} \right) (1 - \alpha^2) p_i^2 + \alpha \left[\left(1 + \frac{1}{N} \right) \alpha - \frac{1}{N} \right] v_i^{(m-1)}.$$

Let $\beta := \alpha \left[\left(1 + \frac{1}{N} \right) \alpha - \frac{1}{N} \right]$. Then for $m \geq 1$,

$$\begin{aligned}v_i^{(m+1)} &= \frac{1}{N} \left[\frac{1 - \beta^{m+1}}{1 - \beta} - \alpha(2 - \alpha) \frac{1 - \beta^m}{1 - \beta} \right] p_i \\ &\quad + \left(1 - \frac{1}{N} \right) \left[\frac{1 - \beta^{m+1}}{1 - \beta} - \alpha^2 \frac{1 - \beta^m}{1 - \beta} \right] p_i^2 \\ &= \frac{\frac{1}{N} p_i}{1 + (1 + 1/N)\alpha} \left[1 + 2\alpha - \left(1 - \frac{1}{N} \right) \alpha \beta^m \right] \\ &\quad + \frac{(1 - \frac{1}{N}) p_i^2}{1 + (1 + 1/N)\alpha} \left[1 + \alpha - \frac{1}{N} \alpha \beta^m \right]\end{aligned}$$

and summing across $i \in [s]$ gives the expression for S_{m+1} .

Note that $0 < \beta < \alpha < 1$, which gives both an upper bound and a lower bound on $v_i^{(m)}$. In particular, for $m \geq 2$

$$v_i^{(m)} < \frac{1}{1 + (1 + 1/N)\alpha} \left[\frac{1}{N} (1 + 2\alpha) p_i + \left(1 - \frac{1}{N} \right) (1 + \alpha) p_i^2 \right]$$

and therefore

$$\begin{aligned}S_m &< \frac{1}{1 + (1 + 1/N)\alpha} \left[\frac{1}{N} (1 + 2\alpha) + \left(1 - \frac{1}{N} \right) (1 + \alpha) S_0 \right] \\ &= 1 - \frac{\left(1 - \frac{1}{N} \right) (1 + \alpha) (1 - S_0)}{1 + (1 + 1/N)\alpha},\end{aligned}$$

where

$$\gamma := \frac{\left(1 - \frac{1}{N} \right) (1 + \alpha) (1 - S_0)}{1 + (1 + 1/N)\alpha} \in \left[\frac{1 + \alpha}{2 + 3\alpha} (1 - S_0), 1 - S_0 \right]$$

since $0 \leq 1/N \leq 1/2$. □

We state a more general result which implies Theorem 3.

Theorem 4. Let $G_n(s)$ and φ be as in Lemma 1. Then for $k \geq 1$ we have

$$\mathbb{E} \|\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)}\|_1 < \frac{1}{N} \sqrt{\frac{n\pi}{\varphi(\zeta)}} G_n(s), \quad (16)$$

where

$$\zeta = \frac{1}{2} - \left(\frac{1}{2} - 2\mathbb{E} \left[\max_{A \subseteq [s]} \mathbf{p}^{(1)}(A) \mathbf{p}^{(1)}([s] \setminus A) \right] \right) \left(\frac{N}{N+n} \right)^2.$$

Proof of Theorem 4. Since all generations share the same original data source, we can write

$$\mathbf{p}^{(k+1)} = \frac{N}{N+n} \mathbf{p}^{(1)} + \frac{n}{N+n} \widehat{\mathbf{p}}^{(k)}, \quad \text{where} \quad \widehat{\mathbf{p}}^{(k)} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i^{(k)} \quad (17)$$

and $\{\mathbf{y}_i^{(k)}\}_{i=1, \dots, n}$ are i.i.d. multinomial with parameter $\mathbf{p}^{(k)}$ and one trial ($k \geq 1$). This gives

$$\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)} = \frac{n}{N+n} [\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(1)}],$$

and applying the triangle inequality yields

$$\|\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)}\|_1 = \frac{n}{N+n} \|\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(1)}\|_1 \leq \frac{n}{N+n} \|\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(k)}\|_1 + \frac{n}{N+n} \|\mathbf{p}^{(k)} - \mathbf{p}^{(1)}\|_1.$$

Taking the expectation and solving the recursion gives for $k \geq 1$

$$\mathbb{E} \|\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)}\|_1 \leq \sum_{j=1}^k \left(\frac{n}{N+n} \right)^{k+1-j} \mathbb{E} \|\widehat{\mathbf{p}}^{(j)} - \mathbf{p}^{(j)}\|_1. \quad (18)$$

From Lemma 1, we have

$$\mathbb{P} \left(\|\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(k)}\|_1 \geq t \mid \mathbf{p}^{(k)} \right) \leq G_n(s) e^{-n\varphi(\pi_k)t^2/4}$$

where $\pi_k := \pi_{\mathbf{p}^{(k)}}$. Integrating over $t \in [0, \infty)$, we get

$$\mathbb{E} \left[\|\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(k)}\|_1 \mid \mathbf{p}^{(k)} \right] \leq G_n(s) \sqrt{\frac{\pi}{n\varphi(\pi_k)}},$$

and by Jensen's inequality and the concavity of $x \mapsto \varphi(x)^{-1/2}$,

$$\mathbb{E} \|\widehat{\mathbf{p}}^{(k)} - \mathbf{p}^{(k)}\|_1 \leq \sqrt{\frac{\pi}{n}} G_n(s) \mathbb{E} [\varphi(\pi_k)^{-1/2}] \leq \sqrt{\frac{\pi}{n\varphi(\mathbb{E}\pi_k)}} G_n(s).$$

Thus

$$\mathbb{E} \|\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)}\|_1 \leq \sqrt{\frac{\pi}{n}} G_n(s) \sum_{j=1}^k \left(\frac{n}{N+n} \right)^{k+1-j} \frac{1}{\sqrt{\varphi(\mathbb{E}\pi_j)}}. \quad (19)$$

It remains to upper bound $\mathbb{E}\pi_j$ since φ is decreasing. For $A \subseteq [s]$ let A^c denote its complement. Then by (17), for $A \subseteq [s]$ we have

$$\begin{aligned} \mathbf{p}^{(k+1)}(A) \mathbf{p}^{(k+1)}(A^c) &= \frac{N^2}{(N+n)^2} \mathbf{p}^{(1)}(A) \mathbf{p}^{(1)}(A^c) + \frac{n^2}{(N+n)^2} \widehat{\mathbf{p}}^{(k)}(A) \widehat{\mathbf{p}}^{(k)}(A^c) \\ &\quad + \frac{Nn}{(N+n)^2} [\mathbf{p}^{(1)}(A^c) \widehat{\mathbf{p}}^{(k)}(A) + \mathbf{p}^{(1)}(A) \widehat{\mathbf{p}}^{(k)}(A^c)] \\ &\leq \frac{N^2}{(N+n)^2} \mathbf{p}^{(1)}(A) \mathbf{p}^{(1)}(A^c) + \frac{n^2}{(N+n)^2} \widehat{\mathbf{p}}^{(k)}(A) \widehat{\mathbf{p}}^{(k)}(A^c) + \frac{Nn}{(N+n)^2}. \end{aligned}$$

Let $\lambda_k := \max_{A \subseteq [s]} \mathbf{p}^{(k)}(A) \mathbf{p}^{(k)}(A^c)$, so the inequality above implies

$$\begin{aligned} \mathbb{E}[\lambda_{k+1} \mid \mathbf{p}^{(1)}, \mathbf{p}^{(k)}] &\leq \frac{N^2}{(N+n)^2} \lambda_1 + \frac{n^2}{(N+n)^2} \mathbb{E} \left[\max_{A \subseteq [s]} \widehat{\mathbf{p}^{(k)}}(A) \widehat{\mathbf{p}^{(k)}}(A^c) \mid \mathbf{p}^{(k)} \right] + \frac{Nn}{(N+n)^2} \\ &\leq \frac{N^2}{(N+n)^2} \lambda_1 + \frac{n^2}{4(N+n)^2} + \frac{Nn}{(N+n)^2} \end{aligned}$$

and thus

$$\mathbb{E}[\lambda_{k+1}] \leq \frac{N^2 \mathbb{E}[\lambda_1] + Nn + n^2/4}{(N+n)^2} = \frac{1}{4} - \left(\frac{1}{4} - \mathbb{E}[\lambda_1] \right) \left(\frac{N}{N+n} \right)^2.$$

Observe that for $0 \leq x \leq 1$ we have $\min(x, 1-x) \leq 2x(1-x)$, so $\mathbb{E}[\pi_k] \leq 2\mathbb{E}[\lambda_k]$. Writing

$$\zeta := \frac{1}{2} - \left(\frac{1}{2} - 2\mathbb{E}[\lambda_1] \right) \left(\frac{N}{N+n} \right)^2$$

we have $\mathbb{E}[\pi_k] \leq \zeta$, so from (19) we see that

$$\mathbb{E} \|\mathbf{p}^{(k+1)} - \mathbf{p}^{(1)}\|_1 \leq \sqrt{\frac{\pi}{n\varphi(\zeta)}} G_n(s) \sum_{j=1}^k \left(\frac{n}{N+n} \right)^{k+1-j} < \frac{1}{N} \sqrt{\frac{n\pi}{\varphi(\zeta)}} G_n(s). \quad (20)$$

□

Proof of Theorem 3. Theorem 3 is an immediate consequence of Theorem 4 by replacing φ with its minimum value 2. □

C Architecture & training parameters for GPT2 experiments

We consider that all generation models have GPT2-type⁴ vanilla architecture which is a decoder-only generative model with the configuration and training parameters as summarized in Table 1. These parameters were chosen to achieve the best validation loss when training the first-generation model on data produced by ground-truth generative model $\mathbf{p}^{(0)}$ as defined in Section 4.1.

Context length	128
Embedding dimension	256
Number of layers	8
Number of self-attention heads	4
Vocabulary size	65
Dropout	0.2
Learning rate	10^{-3}
Batch size	256
Max iterations	2000

Table 1: Architecture and training parameters in the setting of Section 4.1.

⁴<https://github.com/karpathy/nanoGPT>